Types and Sources of Data

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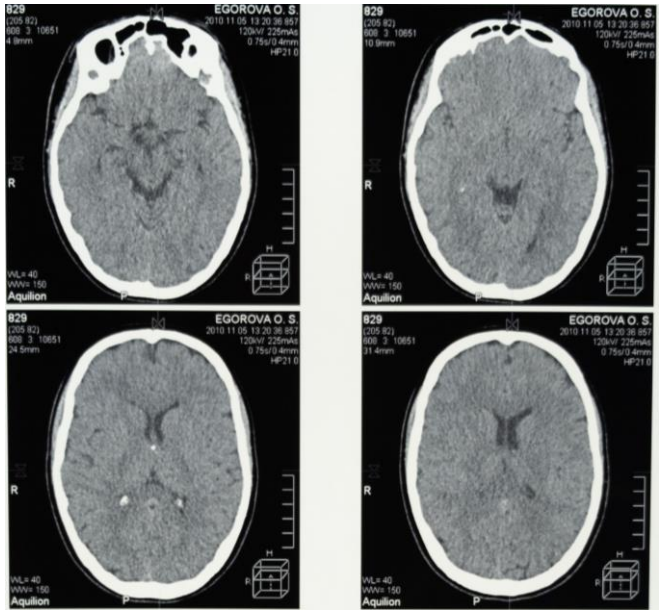
What counts as data?



DATE	TIME	DAY	SBP	DBP
May 8, 2027	7:00 AM	Sunday	129.5 mm Hg	79.5 mm Hg
May 9, 2027	11:00 AM	Monday	120 mm Hg	80 mm Hg
May 10, 2027	8:00 AM	Tuesday	125.5 mm Hg	80.5 mm Hg
May 11, 2027	10:00 AM	Wednesday	115.5 mm Hg	78.5 mm Hg
May 12, 2027	9:00 AM	Thursday	119.5 mm Hg	80.5 mm Hg
May 13, 2027	6:00 AM	Friday	130.5 mm Hg	80.5 mm Hg
May 14, 2027	5:00 AM	Saturday	132 mm Hg	90 mm Hg



Pain:
- ❑ Mild
- ❑ Moderate
- ❑ Severe







Where does data come from?

- Observation
- Experiment
- Population-based study
- Epidemiological data
- Systematic review / meta-analysis

Data does not speak for itself!

Data must be interpreted.

Sources of data



Devices



Laboratory



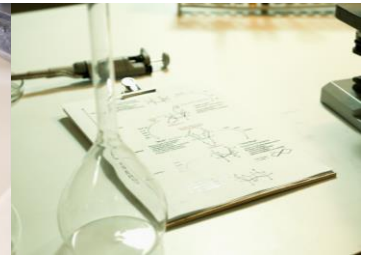
Clinical records



Surveys / questionnaires



Observation



Existing secondary data

Quantitative vs qualitative

Quantitative

Continuous



Numerical data that can take any value within a range.

Discrete



Numerical data that can only take specific, countable values.

Qualitative

Ordinal



Categorical data with a meaningful order or ranking.

Categorical



Data divided into groups or categories with no numerical meaning.

Quantitative often asks “how much / how many?”.

Qualitative often asks “how / why / what does it mean?”

Quantitative vs qualitative

Quantitative

Continuous



Numerical data that can take any value within a range.

Discrete



Numerical data that can only take specific, countable values.

Qualitative

Ordinal



Categorical data with a meaningful order or ranking.

Categorical



Data divided into groups or categories with no numerical meaning.

Exercise: classify each variable:

1. Blood group: A / B / AB / O

2. Pain: none / mild / moderate / severe

3. Body temperature: 37.2°C

4. Number of hospital admissions: 0, 1, 2, 3…

5. Treatment: placebo / drug

6. Hunger ranked from 1 to 10 in severity

Quantitative vs qualitative

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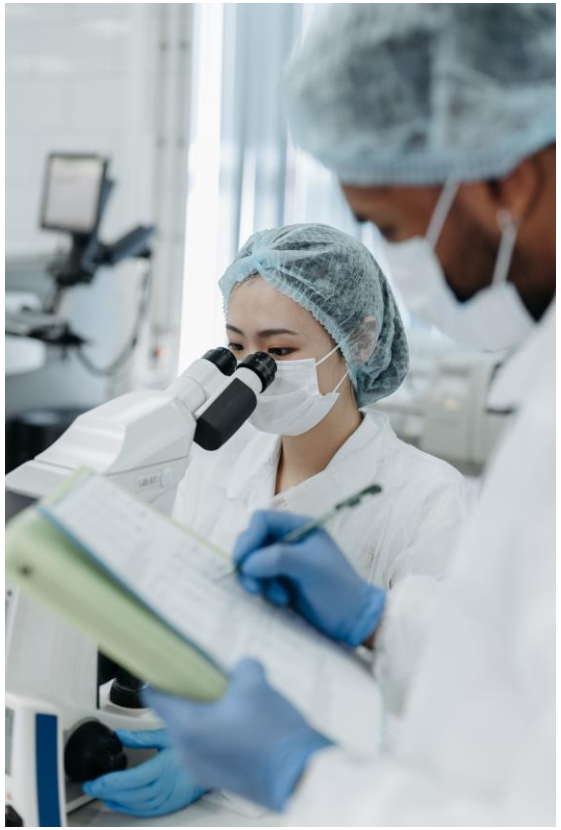


Data divided into groups or categories with no numerical meaning.

Exercise: classify each variable:

1. Eye color
2. Satisfaction: dissatisfied, neutral, satisfied
3. Number of siblings
4. Height: 172.5 cm, 180.2 cm
5. Species of frog
6. Satisfaction: dissatisfied/satisfied

Primary and secondary data



Who collected the data, and for what purpose?



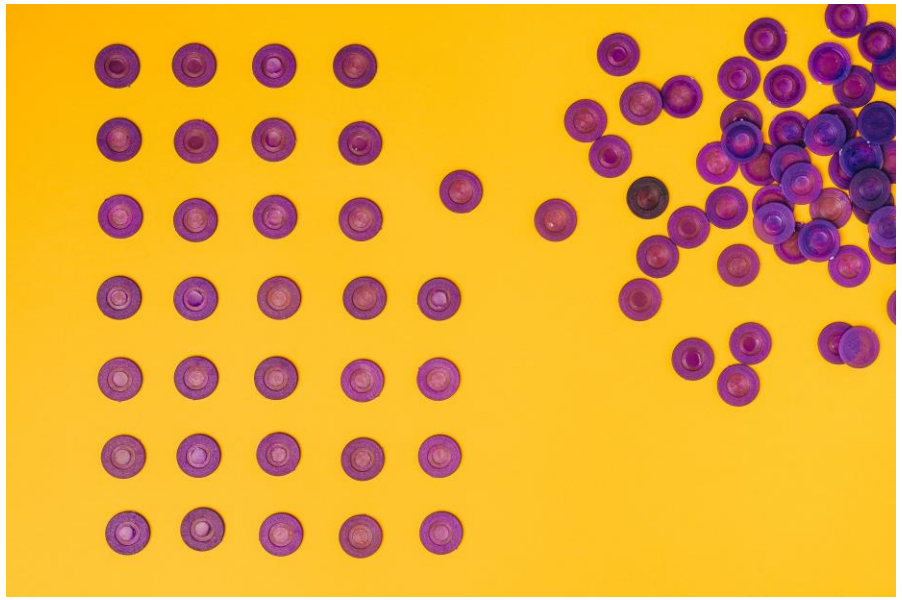
Primary data:
Collected specifically for the research question.
E.g.: A new survey



Secondary data:
Originally collected for another purpose and later used for research.
E.g.: an electronic health-record dataset.

Were the data generated to answer our question or are we adapting existing data?

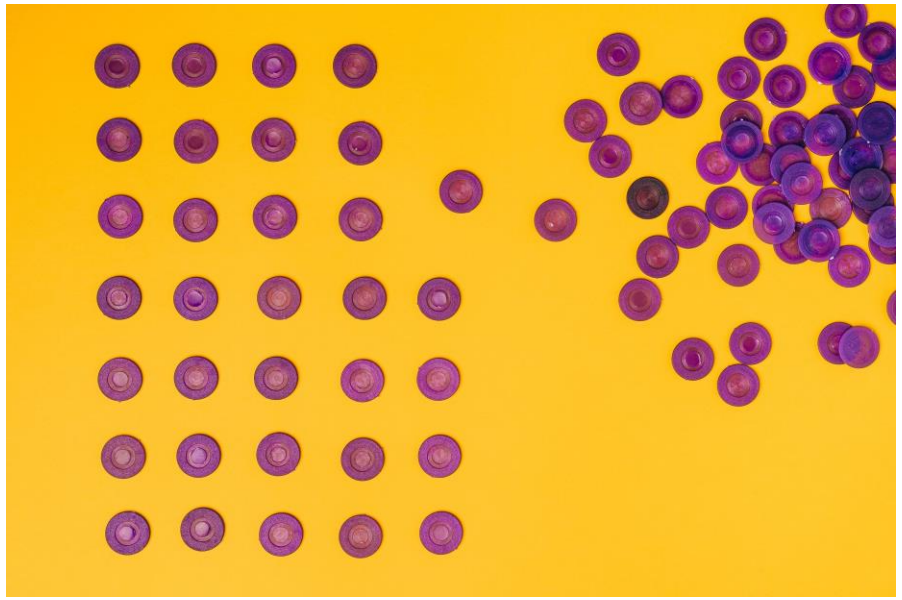
Structured vs unstructured



Structured	Unstructured
Predefined fields and formats; easy to organize in rows/columns.	No fixed tabular format.
Patient ID | Age | BP | Smoker	Interview transcript, clinical free text, medical image, audio or video.

Important: structured ≠ quantitative, and unstructured ≠ qualitative. These are different dimensions.

Structured vs unstructured



Quantitative/qualitative refers to the kind of data.

Structured/unstructured describes the format.

1. Table of daily temperatures
2. CSV of bank transactions
3. Scanned PDF invoice
4. Audio recording of an interview
5. JSON file of sensor readings
6. Free-text customer reviews
7. Free-text customer reviews
8. A picture of an excell spreadsheet

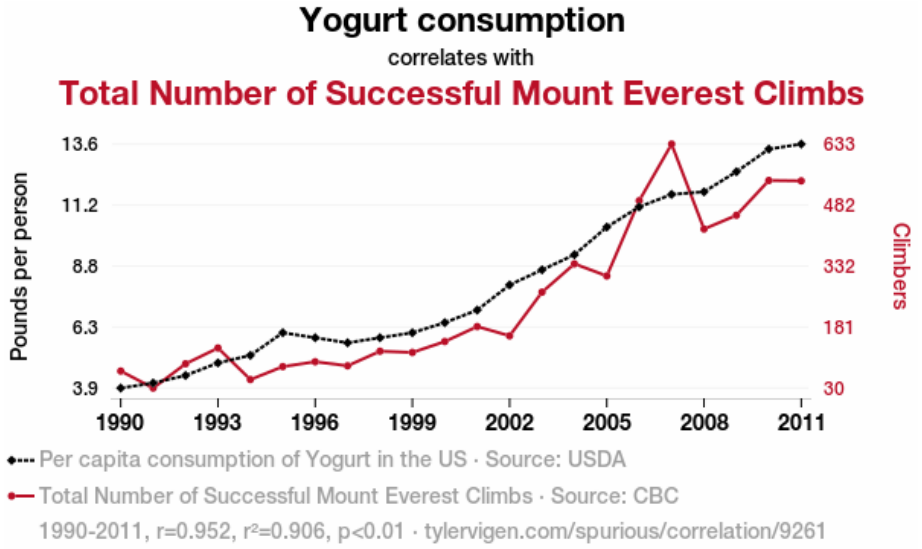
Data relationships

Coincidence: things happen together by chance.

Association: things occur together (e.g. age and body size)

Correlation: Pattern of change

Correlation doesn't mean casuation!





Data quality?



- Is there a clear research question?
- Is the population well defined?
- Were the methods approrpriate?
- Was the follow-up time appropriate?
- Where do scienific data come from?
- Is data of good quality?
- Common problems (biases and confounders)

Good data characteristics



- Complete
- Relevant to the question we are askng
- Consistent (results stable over time and not random)
- Reliable measurements (accurate)

What can go wrong:

- Data can be systematically wrong
- This is called Bias
- Often unintentional

Biases:

Selection bias: wrong people included

Measurement biased: data collected in ways that is not correct or reliable

Publication biased: Most published results are positive results

Confounders



Third factor that influences both factors that we are studying.

Alcohol and lung cancer.

(Does alcohol drinkers have more likelyhood to develop lung cancer?)

Confounder: smoking. People that drink are also more likely to smoke.

Confounders



Third factor that influences both factors that we are studying.

Marihuana consumption and extacy consumption.

(Does marihuana consumption increase the likelyhood to end up consuming extacy?)

Confounder: willingness to break a law. People that are willing to brake a ban for one substance are also more likely to break a ban for another.